QPSK

101 100…

QPSK

2 bit ⟶ 1 symbol

Rician

h·s+n

BER / PSNR

BER ⟶ BER

Rician ⟶

CV

1

2

QPSK

3

AWGN /
Rayleigh /
Rician

4

ħ +

5

BER
PSNR

3 5 1

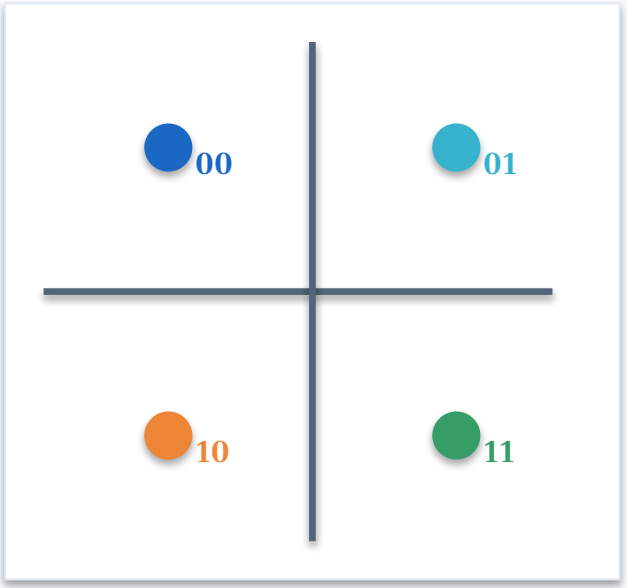
QPSK

bit

00 → 01 → 11 → 10 →

$s = [(2b_1-1) + j(2b_2-1)] / \sqrt{2}$

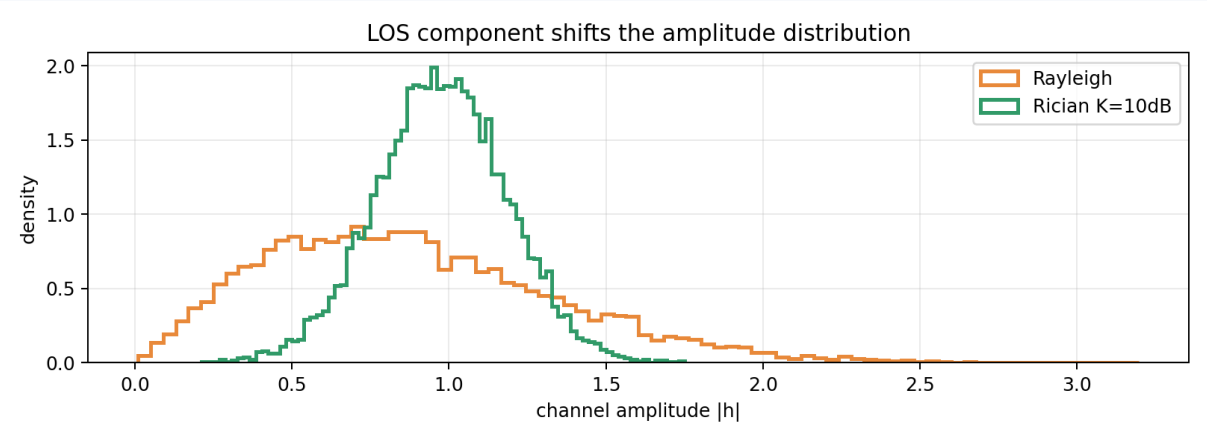
symbol 2 bit
bit Eb/N0



I + jQ

QPSK bit

QPSK



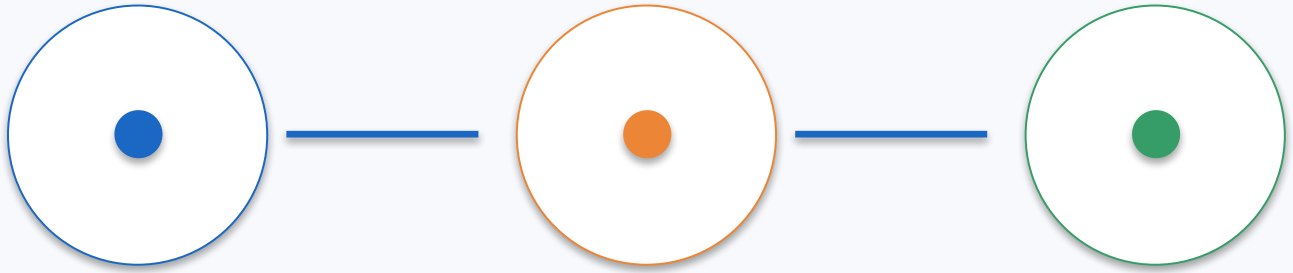
AWGN

h=1

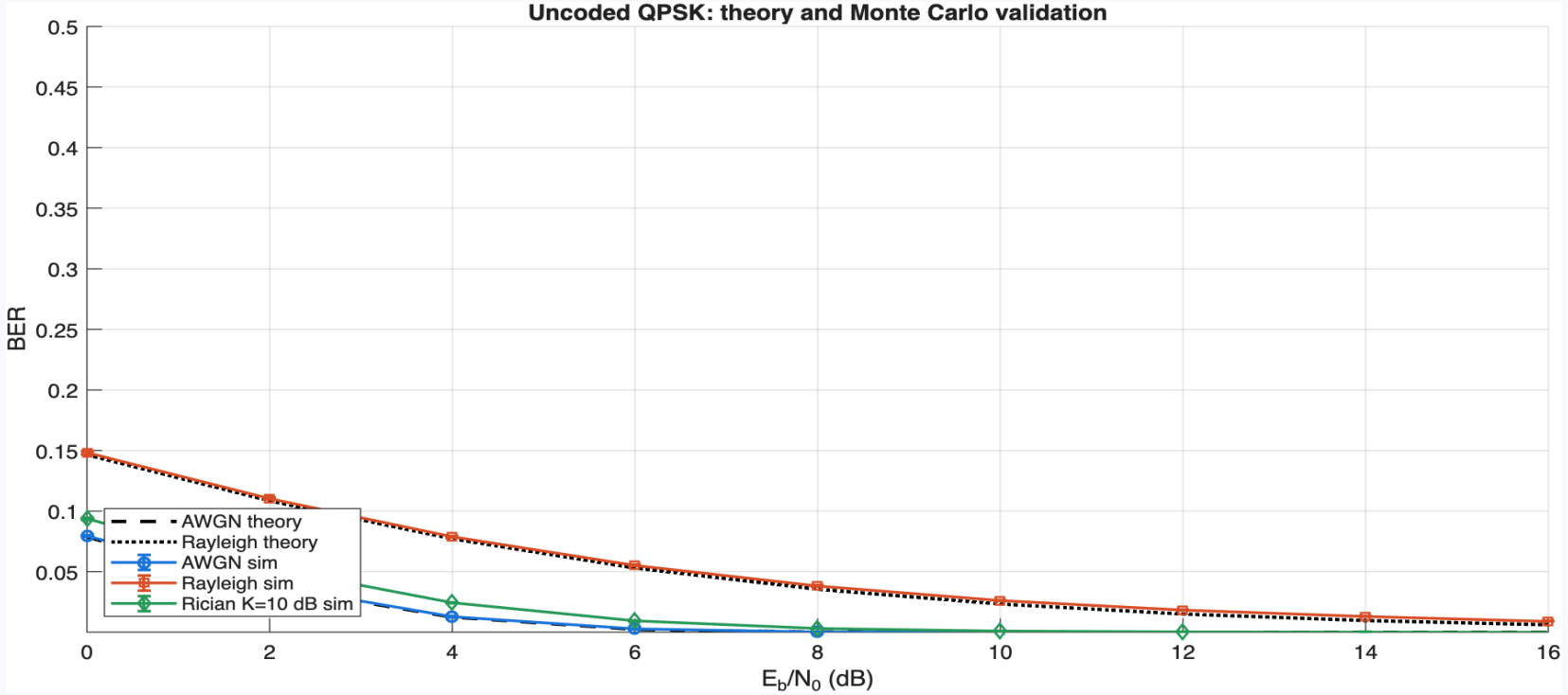
Rayleigh

Rician K=10 dB

h = LOS + diffuse scatter



ħ



AWGN

AWGN

Rayleigh

Rayleigh

Rician

LOS

“ ”

7	3	28	336k
Eb/N0			/

p̂ = bit / bit

CI95 = 1.96 × √[p̂(1−p̂) / N]

CI95

BER 0

“ ”

Viterbi

bit

b_k

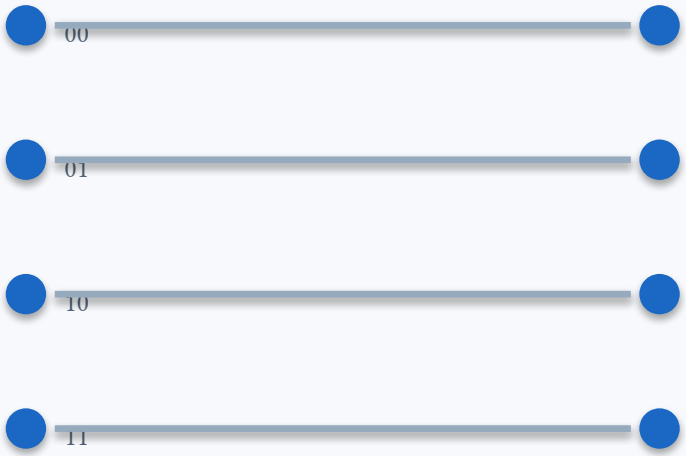
g_1=111
g_2=101

2 bit

Viterbi

4 states

$\hat{b}_k$

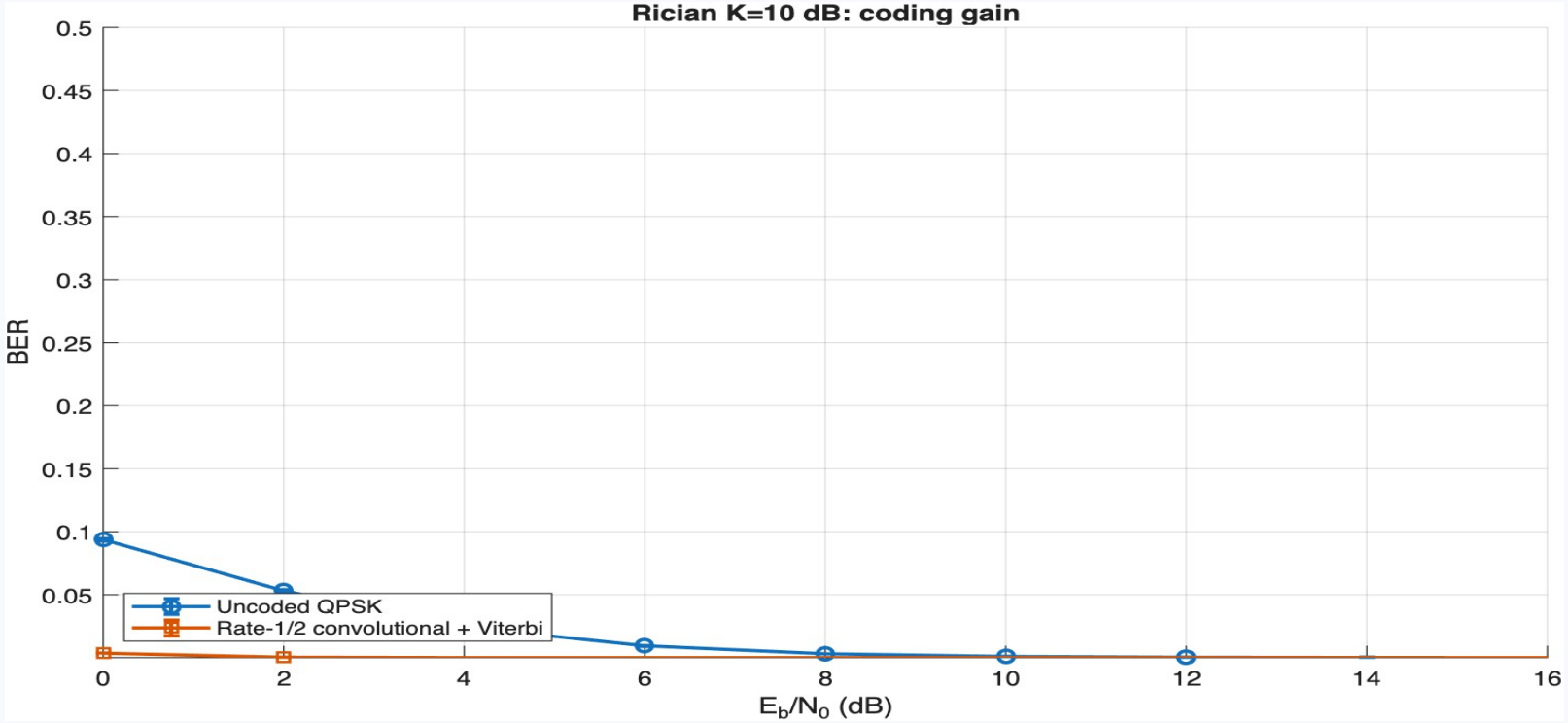


Rician 8 dB

BER ≈ 3.1×10^{-3}

336,000 bit

Rician BER



8 dB 3.1×10^{-3}
BER

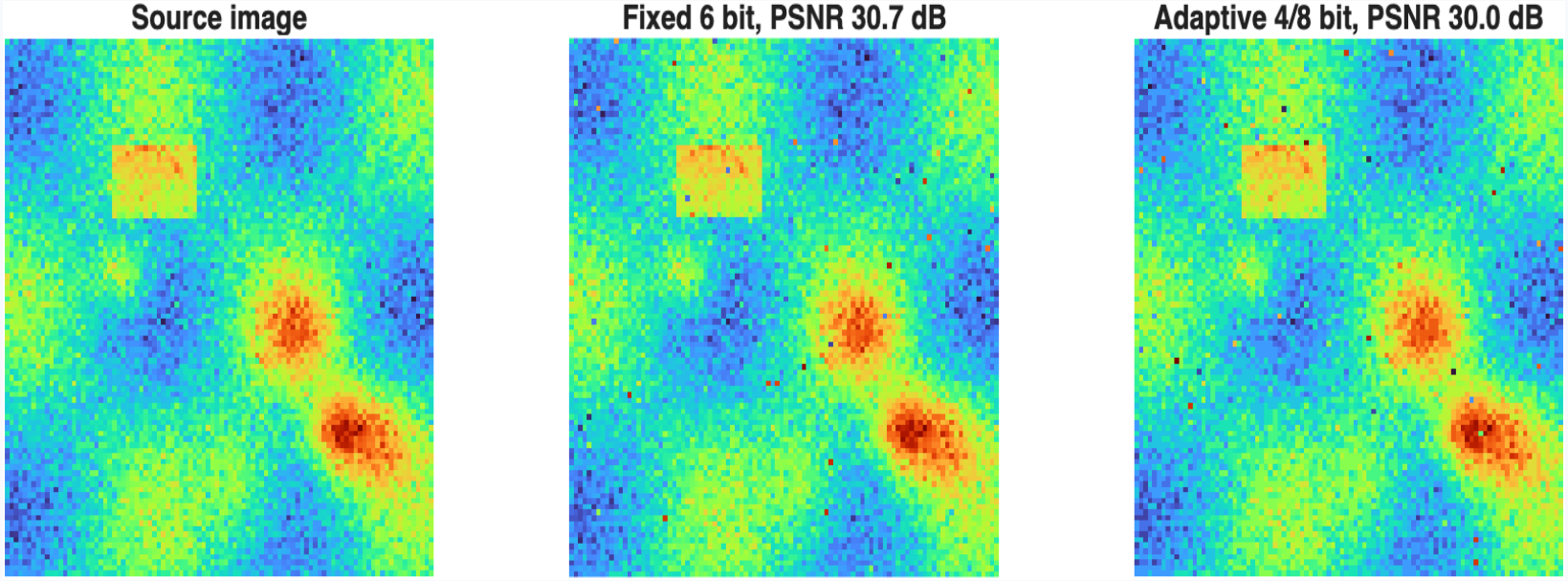
rate-1/2

bit bit
BER

BER
Eb/N0

CV “ ” “ ”

QPSK ✦ Rician



6 bit

CV

4 bit 8 bit

PSNR 30 dB
PSNR 32.3 dB

BER

run_project_v3.m

qpskChannel() / /

convEncode()

viterbiDecode() bit

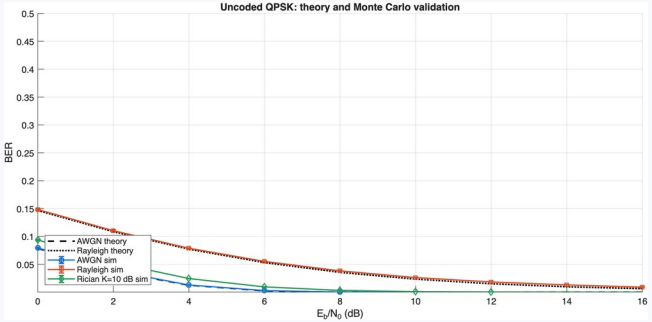
transmitImage()

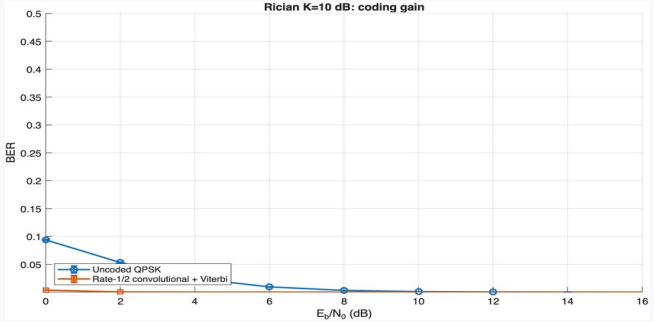
```
h = sqrt(K/(K+1))
  + sqrt(1/(K+1))*g

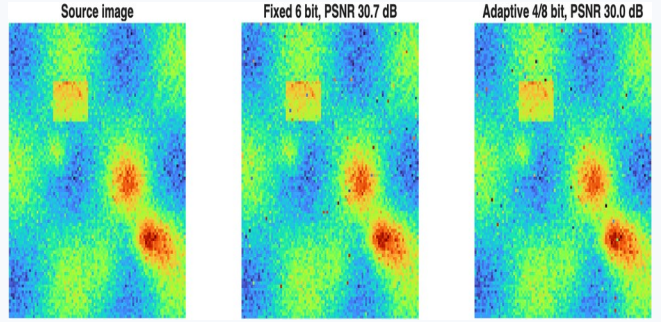
y = h.*s + n
```

Rician

CSV BER







1

2

BER

3

CV

QPSK Rician CV

15

0–3 min

+ QPSK + Eb/N0

3–7 min

AWGN / Rayleigh / Rician +

7–11 min

+ Viterbi + coding gain

11–15 min

CV + + Q&A

“ **BER QPSK Rician** ”

12